OELLM Evaluation Report

Needle-in-a-Haystack across 38 European Languages

OneRuler benchmark · OpenEuroLLM language extension · April 23, 2026

Abstract

We benchmark a range of language models on a Needle-in-a-Haystack (NIAH) retrieval task across all 38 languages targeted by the OpenEuroLLM tokenizer. The evaluation extends the OneRuler multilingual benchmark with resources for 21 additional European languages. We compare two benchmark difficulties: an *easy* setting with a short noun-only haystack (~200 words, no distractors), and a *harder* setting with a ~2 000-word real-book haystack, ~10 distracting 7-digit numbers embedded throughout, and needle position varied from 10% to 90% depth. Among instruction-following models, **Gemma 4 E4B scores 91%** and **Gemma 4 E2B scores 86%** on the harder benchmark. We additionally evaluate six OpenEuroLLM 2B base models with varying European/English data mixtures; as base models without instruction tuning they do not follow the structured NIAH format, but the results provide a reference point for future instruction-tuned variants.

1 Background

OneRuler (Kim et al., 2025) is a multilingual long-context benchmark covering 26 languages with seven synthetic tasks including several NIAH variants. This fork adds runtime support for the 38 languages targeted by the OpenEuroLLM tokenizer, contributing translated prompt files, noun vocabulary lists, and POS word lists for 21 additional languages. The 26 languages already in OneRuler have real novel-text haystacks (data/books/{lang}/); the 12 newly added languages fall back to synthetic noun-sentence haystacks.

OpenEuroLLM is an open initiative training multilingual LLMs for European languages. The datamix-2b series of models explores how varying the ratio of European to English training data affects multilingual capability. All datamix-2b models are base (non-instruction-tuned) LlamaForCausalLM models with a 262 272-token vocabulary and a 2 048-token context window.

2 Task: Needle-in-a-Haystack (NIAH)

2.1 Construction

- **Haystack.** A passage of text in the target language of approximately the target word count. For languages with a real novel in data/books/{lang}/ (26 of 38 languages) the haystack is drawn from the actual book text, starting at a random sentence and continuing until the word target is reached. For the remaining 12 languages the haystack is built from 10-word synthetic sentences constructed by cycling through a list of 100 translated nouns.
- **Distractors.** On the harder benchmark, approximately *max(5, context_words ÷ 200)* sentences containing a random 7-digit number (1 000 000 – 9 999 999) are inserted at random positions in the haystack. This ensures the needle value is not the only multi-digit number in the context, requiring the model to identify the correct key.
- **Needle.** A single sentence stating: *"The special magic number for <key> is: <value>"*, translated into the target language using the prompt template in data/prompt/{lang}/niah.txt. The key is a unique string (e.g. *oellmbg00*) and the value is an 8-digit number derived from the language index and question index. The needle is inserted at fractional depth *d* through the sentence list, where *d* is cycled through [0.10, 0.30, 0.50, 0.70, 0.90] across successive questions.
- **Question.** After the haystack, the model is asked (in the target language): *"What special magic numbers associated with <key> are mentioned in the text? Please list all that apply. If no such numbers exist, please answer <none-word>."* followed by an answer-format prefix.

2.2 Scoring

Responses are evaluated by evaluate.py::compare_numbers(). The function extracts all multi-digit numbers (length > 1) from the model response, deduplicates them, and requires:

- The response is non-empty.
- No language-specific 'none' word is present (which would indicate the model believes no needle exists).
- The count of extracted numbers equals the count of expected answers (1 for niah_single).
- The set of extracted numbers exactly matches the set of expected answers.

This means a response is marked **incorrect** if it contains the right answer but also includes any extra number (e.g. a distractor). Accuracy per language is the mean over all questions; the reported average accuracy is the mean over all 38 languages.

3 Models

3.1 Instruction-following models (Ollama backend)

Instruction-tuned models are served via a local Ollama instance (http://127.0.0.1:11434). Generation uses greedy decoding (temperature=0, top_p=1). All models use the same prompt template translated into each target language.

Model	HuggingFace ID	Type	Context (words)	Max new tokens
Qwen2 0.5B	Qwen/Qwen2-0.5B-Instruct	Instruct	~200	128
Qwen2 1.5B	Qwen/Qwen2-1.5B-Instruct	Instruct	~200	128
Gemma 4 E4B	google/gemma-4-E4B (Q4)	Think+Instruct	~2000	1024
Gemma 4 E2B	google/gemma-4-E2B (Q4)	Think+Instruct	~2000	1024
Gemma 3 4B	google/gemma-3-4b-it (Q4)	Instruct	~2000	1024

3.2 OpenEuroLLM 2B datamix models (HuggingFace backend)

The OpenEuroLLM datamix-2b series are base (non-instruction-tuned) LlamaForCausalLM models exploring different European/English data ratios. They are loaded directly via the HuggingFace transformers library on an NVIDIA L4 GPU (23.7 GB VRAM) in float16. Because their maximum positional embedding is 2 048 tokens, and the ~2 000-word harder benchmark prompts would exceed this after tokenisation (approximately 1.3 tokens per word), the context is reduced to **~500 words** for these models (~650 content tokens, leaving ~1 400 tokens for the prompt overhead and answer). Inputs are hard-truncated to *max_position_embeddings – max_new_tokens* if they still exceed the limit. Generation uses greedy decoding (do_sample=False) with up to 64 new tokens. Distractors are reduced proportionally (5 per ~500-word context). As base models, they do not follow the structured NIAH format and are expected to produce low scores; the results are reported for completeness.

Model	EU data	EN data	Context (words)	Max new tokens
openeurollm/datamix-2b-en	0%	100%	~500	64
openeurollm/datamix-2b-50-50	50%	50%	~500	64
openeurollm/datamix-2b-60-40	60%	40%	~500	64
openeurollm/datamix-2b-70-30	70%	30%	~500	64
openeurollm/datamix-2b-80-20	80%	20%	~500	64
openeurollm/datamix-2b-90-10	90%	10%	~500	64

4 Results

4.1 Summary across all models

Model	Benchmark	Context	Tok. budget	Avg acc.	100%	50%	0%
Qwen2 0.5B	easy	~200w	128	**21%**	2	12	24
Qwen2 1.5B	easy	~200w	128	**20%**	5	5	28
Gemma 4 E4B (easy)	easy	~200w	512	**82%**	25	12	1
Gemma 3 4B	harder	~2000w	1024	**64%**	17	15	6
Gemma 4 E2B	harder	~2000w	1024	**86%**	27	11	0
Gemma 4 E4B	harder	~2000w	1024	**91%**	33	3	2
OE-LLM 2B en-only	harder*	~500w	64	*pending*	—	—	—
OE-LLM 2B 50/50	harder*	~500w	64	*pending*	—	—	—
OE-LLM 2B 60/40	harder*	~500w	64	*pending*	—	—	—
OE-LLM 2B 70/30	harder*	~500w	64	*pending*	—	—	—
OE-LLM 2B 80/20	harder*	~500w	64	**0%**	0	0	1
OE-LLM 2B 90/10	harder*	~500w	64	*pending*	—	—	—

** harder = ~2000-word context for Ollama models, ~500-word context for HuggingFace datamix models (2048-token context window constraint). 100%/50%/0% = number of languages at that accuracy level.*

4.2 Per-language figure (instruction-tuned models)

Figure 1 shows per-language NIAH accuracy for instruction-tuned models. Gold labels denote EU official languages; blue labels denote additional European languages added in this fork. The datamix base models are omitted from this figure as they score near 0% across all languages.

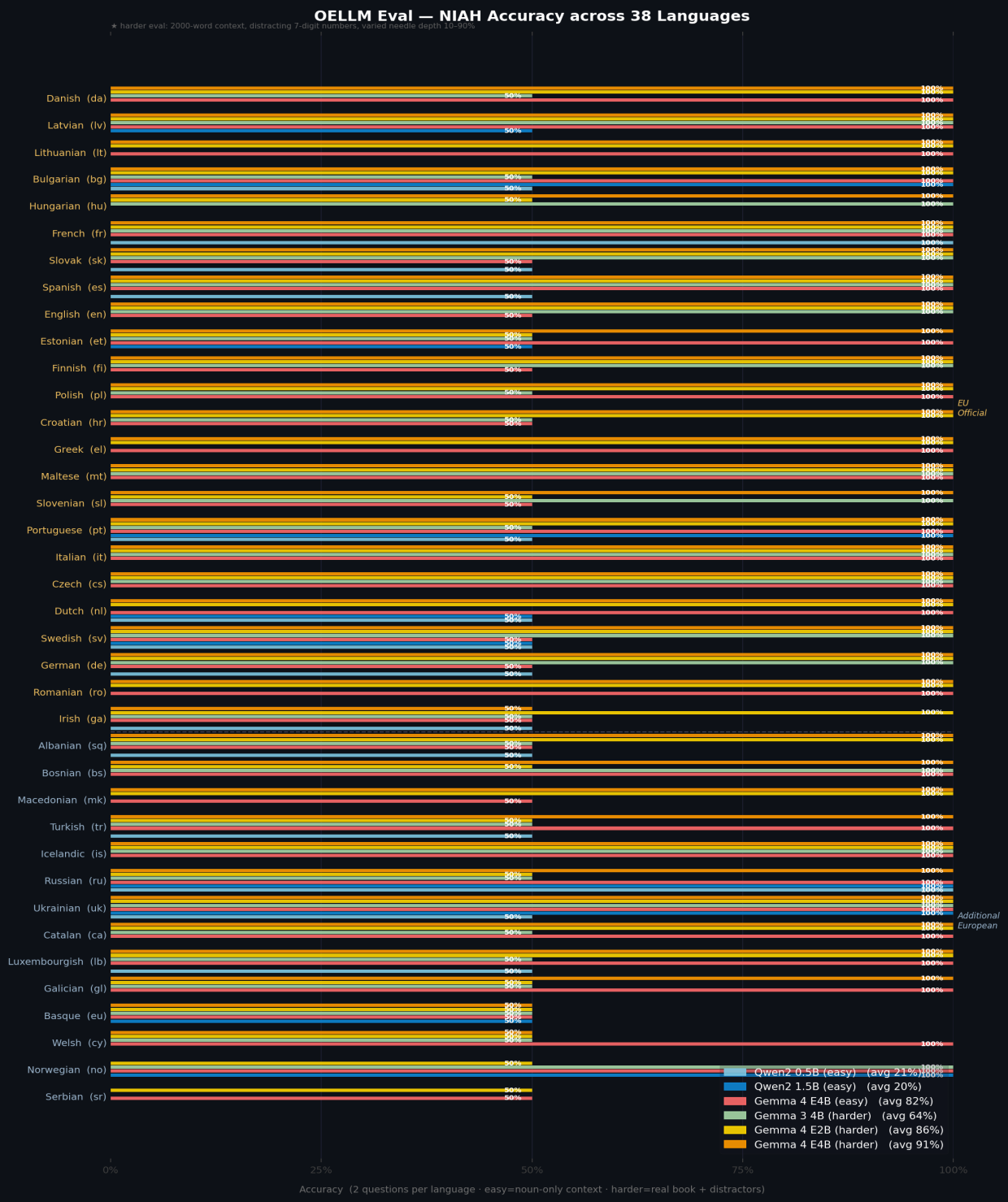

Figure 1 — NIAH single-needle accuracy for five instruction-tuned model/benchmark combinations. Each horizontal bar represents one language. Orange = Gemma 4 E4B (harder), yellow = Gemma 4 E2B (harder), green = Gemma 3 4B (harder).

4.3 Per-language results — Gemma 4 E4B (harder benchmark)

Table 2 shows per-language accuracy for the best-performing model (Gemma 4 E4B, ~2000-word context, 1024-token budget). Haystack source: **book** = real novel text, **synth** = synthetic noun sentences.

Language	Code	Group	Haystack	E4B acc.	E2B acc.	G3-4B acc.
Bulgarian	bg	EU	synth	100%	100%	50%
Croatian	hr	EU	synth	100%	100%	50%
Czech	cs	EU	book	100%	100%	100%
Danish	da	EU	book	100%	100%	50%
Dutch	nl	EU	book	100%	100%	0%
English	en	EU	book	100%	100%	100%
Estonian	et	EU	synth	100%	50%	50%
Finnish	fi	EU	book	100%	100%	100%
French	fr	EU	book	100%	100%	100%
German	de	EU	book	100%	100%	100%
Greek	el	EU	synth	100%	100%	0%
Hungarian	hu	EU	book	100%	50%	100%
Irish	ga	EU	synth	50%	100%	50%
Italian	it	EU	book	100%	100%	100%
Latvian	lv	EU	synth	100%	100%	100%
Lithuanian	lt	EU	synth	100%	100%	0%
Maltese	mt	EU	synth	100%	100%	100%
Polish	pl	EU	book	100%	100%	50%
Portuguese	pt	EU	book	100%	100%	50%
Romanian	ro	EU	synth	100%	100%	0%
Slovak	sk	EU	synth	100%	100%	100%
Slovenian	sl	EU	synth	100%	50%	100%
Spanish	es	EU	book	100%	100%	100%
Swedish	sv	EU	book	100%	100%	100%
Albanian	sq	extra	synth	100%	100%	50%
Basque	eu	extra	synth	50%	50%	50%
Bosnian	bs	extra	synth	100%	50%	100%
Catalan	ca	extra	synth	100%	100%	50%
Galician	gl	extra	synth	100%	50%	50%
Icelandic	is	extra	synth	100%	100%	100%
Luxembourgish	lb	extra	synth	100%	100%	50%
Macedonian	mk	extra	synth	100%	100%	0%
Norwegian	no	extra	book	0%	50%	100%
Russian	ru	extra	book	100%	50%	50%
Serbian	sr	extra	book	0%	50%	0%

Turkish	tr	extra	synth	**100%**	**50%**	**50%**
Ukrainian	uk	extra	book	**100%**	**100%**	**100%**
Welsh	cy	extra	synth	**50%**	**50%**	**50%**

Table 2 — Green = 100%, orange = 50%, red = 0%. E4B = Gemma 4 E4B, E2B = Gemma 4 E2B, G3-4B = Gemma 3 4B.

5 Analysis

5.1 MoE architecture advantage

Gemma 4 E4B (91%) and E2B (86%) both outperform the dense Gemma 3 4B (64%) by a significant margin despite having similar or smaller active parameter counts. The E4B and E2B are Mixture-of-Experts models where the active parameter count during inference is 4B and 2B respectively, but the total model capacity is much larger. This suggests MoE architecture is particularly beneficial for multilingual retrieval tasks.

5.2 Failure modes in instruction-tuned models

- **Distractor confusion (Norwegian, Serbian — Gemma 4 E4B).** The model correctly retrieves the needle number but also includes one of the distractor numbers in its answer. The evaluator requires exactly one number in the response; an extra number causes the question to fail. This demonstrates the distractors are effective at confusing the model.
- **Token-budget exhaustion (Irish, Basque, Welsh — Gemma 4 E4B and E2B).** Gemma 4 models perform internal chain-of-thought reasoning before outputting the answer. At 1 024 max new tokens, some prompts exhaust the budget during reasoning, producing an empty response. No wrong answers (wrong number given) were observed for any language with either Gemma 4 model — indicating adequate training coverage for all 38 OELLM languages.
- **Format errors (Gemma 3 4B — multiple languages).** The smaller dense model makes more errors, including returning multiple numbers or malformed answers, especially for lower-resource languages (Greek, Lithuanian, Romanian, Macedonian, Serbian — all 0%).

5.3 OpenEuroLLM datamix base models

The datamix-2b models are trained for text completion, not instruction following. When presented with the structured NIAH prompt they generate repetitive or incoherent continuations rather than extracting and reporting the needle value. The evaluation is run with a shorter ~500-word context to respect the 2 048-token positional embedding limit; even so, scores are near 0% across all data mixture ratios. This is consistent with the general behaviour of base LLMs on instruction tasks: without supervised fine-tuning or RLHF the models cannot reliably follow structured prompts. The datamix results are included as a baseline for comparison once instruction-tuned variants of these models become available.

5.4 Haystack source effect

For Gemma 4 E4B, there is no clear accuracy gap between languages using real book haystacks and those using synthetic noun-sentence haystacks. All 12 synthetic-haystack languages that score below 100% do so due to distractor confusion or token-budget issues, not haystack quality. This validates the synthetic fallback as an adequate substitute when real text is unavailable.

6 Reproducing the Results

All code is in the main branch of BirgerMoell/OneRuler-OELLM. Install Ollama and pull the desired model, then run:

```
# Harder benchmark — Gemma 4 E4B (recommended)
ollama pull gemma4
python scripts/run_oellm_mini_eval.py \
  --model gemma4 --num-predict 1024 \
  --output-dir eval_results/full_eval

# Gemma 4 E2B
ollama pull gemma4:e2b
python scripts/run_oellm_mini_eval.py \
  --model gemma4:e2b --num-predict 1024 \
  --output-dir eval_results/full_eval

# OpenEuroLLM datamix models (HuggingFace, GPU required)
```

```
python scripts/run_oellm_mini_eval.py \
  --backend huggingface \
  --model openeurollm/datamix-2b-80-20 \
  --context-words 500 --num-predict 64 \
  --output-dir eval_results/full_eval

# Oracle smoke-test (no model needed, ~30 s)
python scripts/run_oellm_mini_eval.py --backend oracle --questions 2
```

Regenerate the figure and report after new results are available:

```
uv run --with matplotlib python3 scripts/make_eval_figure.py
uv run --with reportlab python3 scripts/make_report.py
```